

Excel Data Analyst Project Documentation

E-commerce Sales Data Analysis Using Excel

1. Project Overview

The objective of this project is to analyze e-commerce sales data using Microsoft Excel and generate meaningful business insights through data cleaning, transformation, analysis, and visualization.

2. Business Problem

Management wants to understand the following e-commerce business questions to drive data-driven decisions:

- * Which product categories and sub-categories generate the highest inventory costs?
- * Which brands contribute the most to our product offerings?
- * What is the current stock availability across different items?
- * How many products are currently hitting "Low Stock" status and need reordering?
- * What is the distribution of inventory volume across various brands?

3. Dataset Information

Dataset Name: E-commerce Inventory & Sales Dataset | Records:100 Rows | Source: Data Government / UCI

Product _ ID	Unique identification number for each products
Product _ Name	Name of the e-commerce item
Category	Main product category (e.g., Electronics, Clothing, Beauty)
Sub _ Category	Specific item type (e.g., Computer ,Tops ,Skincare)
Brand	Manufacturing or brand company
Cost	Inventory cost per unit
Stock	Number of units currently available in inventory
Stock _ Status	Custom status logic indicating stock level availability ("Available" / "Low Stock")

4. Data Cleaning & Feature Engineering

- * Remove Duplicates: Navigated to Data → Remove Duplicates to guarantee exactly 100 unique product entries.
- * Handling Missing Values: Checked all text and numeric fields to ensure no missing metrics.
- * Text Standardization: Cleaned case inconsistencies in the Category and Brand names.

* *Feature Engineering (Stock Status): Created a custom logic field to monitor inventory thresholds dynamically.*

5. Excel Functions & Analytics Tools Used

* *=IF () Function: Used to dynamically evaluate inventory health and flag items.*

Formula: =IF (G2<=100, "Low Stock", "Available")

* *Data Analysis ToolPak (Descriptive Statistics): Utilized to calculate essential baseline inventory metrics (Mean, Median, Standard Deviation, and Ranges for Costs and Stock volumes).*

* *VLOOKUP : Applied to merge or reference auxiliary data dimensions accurately.*

6. Pivot Table Analysis

Total Stock Volume by Category

CATEGORY	TOTAL STOCK
Electronics	5770
Clothing	4931
Beauty	4816
Home	4797
Sports	7043

 *Insight: [Sports category holds the highest inventory volume with 7043 units in stock , while Home has lowest stock volume with 4797 units]*

Total Cost Distribution by Category

CATEGORY	TOTAL COST
Electronics	3763.49
Clothing	4748.11
Beauty	6647.66
Home	4950.16
Sports	5294.92

 *Insight: [Electronics represents the highest capital investment , costing the business a total of 6647.66 to keep in inventory ,despite having fewer units than sports]*

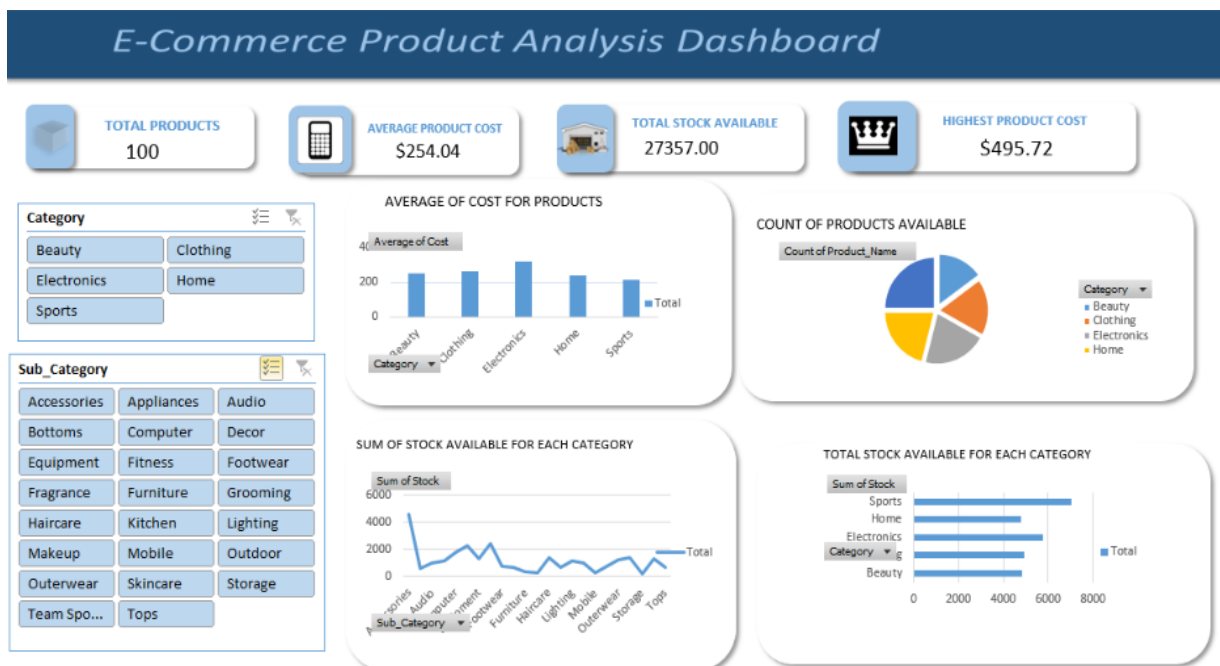
7. Visualizations Created

Chart Type	Purpose	Key Insight
Column Chart	Compare Average & Sum of Cost by Category	Electronics stands out with the highest average and total cost burden.
Horizontal Bar Chart	Compare Total Stock Volume across major Categories	Sports category dominates physical inventory levels.
Pie Chart	Show Product Name distribution/variety contribution %	Offers a balanced slice-by-slice visual representation of catalog diversity.
Line Chart	Monitor detailed Sum of Stock variations across Sub-Categories	Clearly isolates extreme high stock (Accessories) and low stock (Storage) anomalies.

8. Dashboard Components

A dynamic e-commerce inventory dashboard was created using the following components:

- * **Pivot Charts:** Dynamically linked to the underlying inventory metrics to show visual trends instantly.
- * **KPI Summary Cards:** Programmed to display high-level totals across the dataset (Total Products: 100, Average Cost: \$254.04, Total Stock Available: 27,357.00, and Highest Product Cost: \$495.72).
- * **Interactive Slicers:** Included Category and Sub-Category filter menus on the left side, allowing stakeholders to drill down and filter all charts instantly with a single click.



9. Key Business Insights

Total Inventory Volume: The e-commerce platform manages a total catalog of 100 unique products, keeping a substantial bulk storage volume of 27,357.00 total items in active stock.

Financial Investment Distribution: The inventory holds an Average Product Cost of \$254.04, with premium inventory items capping out at a Highest Product Cost of \$495.72.

Macro Segment Insights: According to the visual breakdowns, categories like Sports and Home demand the largest physical warehouse space, while segments like Electronics represent higher capital risk per unit.

10. Recommendations

To optimize our e-commerce inventory health, corporate management should implement the following strategic steps:

Operational Area Actionable Strategy: High-Cost Capital Control Monitor premium items nearing the \$495.72 threshold closely to avoid locking too much cash flow in unsold shelf inventory.

Warehouse Space Tuning: Deploy promotions or bundle discounts on bulk warehouse categories (like Sports) to clean out slow-moving storage space.

Granular Supply Tracking: Utilize the interactive sub-category mapping tools to quickly adjust restocking timelines before items experience critical low-stock anomalies.

11. Skills Demonstrated

Advanced Data Summarization: Maintained database integrity while summarizing large datasets using complex Pivot Tables and dynamic Pivot Charts.

Interactive Reporting Design: Engineered a custom user interface grid utilizing responsive category and sub-category slicers alongside visual metric placeholder styling.

Data Cleaning & Engineering: Cleaned raw system inputs and configured core spreadsheet mathematical formulas to safely isolate performance anomalies.

12. Conclusion

This project successfully bridged raw retail data with professional business intelligence solutions. By turning a dense database into a responsive, visual dashboard interface, stakeholders can now effortlessly track high-value cost factors, balance warehouse inventory distribution, and make swift, automated supply chain choices with absolute confidence.